Master's Thesis at the Pattern Recognition Lab, FAU Erlangen-Nuremberg

Alignment-Stable Attention for Writer-Independent IMU Handwriting Recognition via Hybrid CTC–AR Training, Calibrated Decoding, and Pretrained LM Decoder Adaptation

Online handwriting recognition (HWR) from inertial measurement units (IMUs) enables writing without cameras, touchscreens, or specialized digitizers, making it attractive for privacy-preserving and low-infrastructure text input. In the writer-independent (WI) setting, recognition performance is limited by substantial inter-writer variability (e.g., stroke dynamics, writing speed, grip and sensor placement), sensor noise, and distribution shift between training and unseen data. CNN-based encoders with connectionist temporal classification (CTC) decoding provide a strong baseline for WI IMU-based HWR, yet errors are still driven by imperfect alignment and decoding decisions, which can manifest as character-level substitutions as well as word-level mistakes. This thesis builds on the REWI baseline [1] and our current attention-based pipeline to improve both character error rate (CER) and word error rate (WER) across word-level and sentence-level datasets under explicit memory and latency constraints. Building on our current AR pipeline, we observe a clear trade-off: auto-regressive (AR) decoding substantially improves WER, while CER can degrade moderately due to imperfect character-level alignment. The primary objective of this thesis is therefore to achieve simultaneous improvements in both CER and WER by adopting a hybrid training regime that combines CTC and AR supervision: CTC provides strong monotonic, character-level alignment supervision, while AR training leverages global context to reduce word-level errors. To further reduce residual errors, we will stabilize attention alignment and decoding behavior through alignment-aware regularization (e.g., skip/coverage penalties) and calibrated beam-search decoding (length normalization and EOS control), and we will optionally integrate a lightweight external language model for fusion/rescoring during beam search if sufficient transcripts are available. As a secondary objective (time permitting), we will explore a multimodal variant that retains the same CNN feature extractor while conditioning a pretrained language model decoder on the encoder tokens and fine-tuning it for IMU-HWR, in order to assess whether pretrained decoders can match or exceed the hybrid approach under comparable accuracy and efficiency constraints.

The thesis consists of the following milestones:

- **Baseline reproduction and protocol.** Reproduce the current AR pipeline on WI splits with consistent CER/WER reporting; configurations, and evaluation scripts.
- **Hybrid multitask learning (CTC+AR).** Add a CTC auxiliary head on encoder outputs and train with a joint objective (AR cross-entropy + λ·CTC) [2]. Perform a controlled sweep over λ and regularization to stabilize alignment.
- **Decoding study (calibrated beam search).** Implement beam search for AR decoding; tune length normalization and EOS control (EOS bias and/or minimum-length constraints). Add an alignment-aware *training-time* regularizer on decoder cross-attention (coverage/skip penalty) and evaluate *decoding-time* heuristics (coverage penalty and/or length-normalized scoring in beam search). Quantify improvements on WI validation with CER/WER and tail-risk metrics.
- **Evaluation and ablations.** Primary: CER/WER on WI test writers. Secondary: tail-risk metrics (distribution of normalized edit distance), writer-stratified performance, and confusion/collision

analysis. Ablate (i) loss regime (CTC/AR/joint), (ii) decoding settings, (iii) LM-decoder integration choices, and (iv) masked pretraining, where applicable.

- **Second-priority extension: pretrained LM decoder (multimodal).** After establishing a strong hybrid CTC+AR baseline, integrate a pretrained seq2seq LM decoder and condition it on the CNN encoder's downsampled IMU tokens via a lightweight projection/adapter. Fine-tune under a strict parameter/latency budget, and compare against the scratch AR decoder. Ground the LM decoding with the auxiliary CTC signal to mitigate fluent-but-incorrect outputs.

- **Generalization via masked modeling (optional if time allows).** Pretrain the existing CNN encoder using self-supervised masking (time-span/channel masking) and a reconstruction or latent-prediction objective [3]. Fine-tune on supervised HWR and measure WI robustness gains.

- **Encoder adaptor study (optional if time allows).** Evaluate a lightweight encoder adaptor by inserting a small Transformer encoder (2–4 layers) on the downsampled CNN tokens before the AR/LM decoder, under a fixed budget [4]. Run controlled ablations for AR-only vs hybrid training.

The implementation should be done in Python.

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References

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[3] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

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